Time Reversal Test

It is an important test for testing the consistency of a good index number. This test maintains time consistency by working both forward and backward with respect to time (here time refers to base year and current year). Symbolically the following relationship should be satisfied, $P_{01} \times P_{10} = 1$

Fisher's index number formula satisfies the above relationship

$$P_{01}^{F} = \sqrt{\frac{\sum p_1 q_0 \times \sum p_1 q_1}{\sum p_0 q_0 \times \sum p_0 q_1}}$$

when the base year and current year are interchanged, we get

$$P_{10}^{F} = \sqrt{\frac{\sum p_0 q_1 \times \sum p_0 q_0}{\sum p_1 q_1 \times \sum p_1 q_0}}$$

$$P_{01}^{F} \times P_{10}^{F} = 1$$

Note

Ignore the factor 100 in each Index number

Factor Reversal Test

This is another test for testing the consistency of a good index number. The product of price index number and quantity index number from the base year to the current year should be equal to the true value ratio. That is, the ratio between the total value of current period and total

value of the base period is known as true value ratio. Factor Reversal Test is given by,

$$P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

where $P_{01} = \sqrt{\frac{\sum p_1 q_0 \times \sum p_1 q_1}{\sum p_0 q_0 \times \sum p_0 q_1}}$

Now interchanging P by Q, we get

$$Q_{01} = \sqrt{\frac{\sum q_1 p_0 \times \sum q_1 p_1}{\sum q_0 p_0 \times \sum q_0 p_1}}$$

where P_{01} is the relative change in price.

Q_{01} is the relative change in quantity.

Example 9.12

Calculate Fisher's price index number and show that it satisfies both Time Reversal Test and Factor Reversal Test for data given below.

DO YOU KNOW?

Since Fisher's Price Index Number satisfies both TRT & FRT, it is termed as an Ideal Index Number

Commodities	Price		Quantity	
	2003	2009	2003	2009
Rice	10	13	4	6
Wheat	15	18	7	8
Rent	25	29	5	9
Fuel	11	14	8	10
Miscellaneous	14	17	6	7

Solution:

Commodities	Price		Quantity		p_0q_0	p_0q_1	p_1q_0	p_1q_1
	2003 (p_0)	2009 (p_1)	2003 (q_0)	2009 (q_1)				
Rice	10	13	4	6	40	60	52	78
Wheat	15	18	7	8	105	120	126	144
Rent	25	29	5	9	125	225	145	261
Fuel	11	14	8	10	88	110	112	140
Miscellaneous	14	17	6	7	84	98	102	119
				Total	442	613	537	742

Table 9.11

Fisher's price index number

$$P_{01}^{F} = \left(\sqrt{\frac{\sum p_1 q_0 \times \sum p_1 q_1}{\sum p_0 q_0 \times \sum p_0 q_1}} \right) \times 100$$

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